

Proof. Pointwise bound. Define the indicator of the event $\{X \geq a\}$ by

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

For every ω we have

$$a \mathbf{1}_{\{X \geq a\}}(\omega) \leq X(\omega),$$

and dividing by the positive constant a yields the pointwise inequality

$$\mathbf{1}_{\{X \geq a\}} \leq \frac{X}{a} \quad \text{a.s.} \quad (1)$$

Take expectations. Both sides of (1) are non-negative, so by monotonicity of expectation,

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \mathbb{E}\left[\frac{X}{a}\right]. \quad (1)$$

Using linearity of expectation,

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \frac{\mathbb{E}[X]}{a}. \quad (2)$$

Replace the expectation of the indicator. For any event A , $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. Hence

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = \mathbb{P}(X \geq a). \quad (3)$$

Substituting (3) into (2) gives the desired inequality

$$\boxed{\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}}. \quad (2)$$

Edge (degenerate) cases.

- If $\mathbb{E}[X] = 0$, then $X = 0$ almost surely, so $\mathbb{P}(X \geq a) = 0$ for every $a > 0$, and the inequality holds with equality.
- If $a \downarrow 0$, the hypothesis $a > 0$ prevents division by zero; letting $a \rightarrow 0^+$ makes the right-hand side diverge, so the bound remains true (vacuously).

Equality condition. Equality in the final inequality can occur only if every intermediate inequality is an equality almost surely. Equality in (1) requires

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \frac{X(\omega)}{a} \quad \text{for a.e. } \omega,$$

i.e. $X(\omega) = a \mathbf{1}_{\{X \geq a\}}(\omega)$. Hence X takes only the values 0 and a almost surely. Thus

$$\mathbb{P}(X \in \{0, a\}) = 1 \quad \Longleftrightarrow \quad X = a \mathbf{1}_{\{X \geq a\}} \quad \text{a.s.} \quad (4)$$

and this condition is both necessary and sufficient for equality.

Conclusion. Using the pointwise bound $a \mathbf{1}_{\{X \geq a\}} \leq X$, monotonicity and linearity of expectation, and $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$, we have derived Markov's inequality together with a complete description of edge cases and the equality scenario.

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